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Experimentation in Software Engineering

DAT246 Empirical Software Engineering



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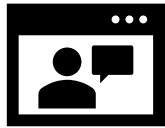
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Disambiguation

Experiments in software engineering research are strongly associated with the **causal effect of a treatment** *compared to a baseline*, for example:



Do developers produce
better code faster when
using **ChatGPT**?



Do students learn
programming *easier* when
using **Python or Java**?



Are requirements written
with **UML** *easier to test*
than **natural language**?

Experimentation is typically *not* referring to simply “trying things out” in software engineering research.



Definitions

Concepts and Terms relevant to Experimentation

Definition

A controlled experiment is an empirical research method where we randomly assign a treatment to combinations of subjects and objects and observe its effect on one or more dependent variables.

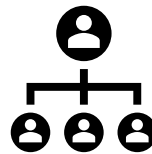
Definition: Subjects

A controlled experiment is an empirical research method where we randomly assign a treatment to combinations of **subjects** and objects and observe its effect on one or more dependent variables.

Subject: unit of analysis



Person



Organization

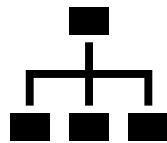


Software

Definition: Objects

A controlled experiment is an empirical research method where we randomly assign a treatment to combinations of subjects and **objects** and observe its effect on one or more dependent variables.

Object: unit to which the treatment will be applied



UML Model



Specification



Software

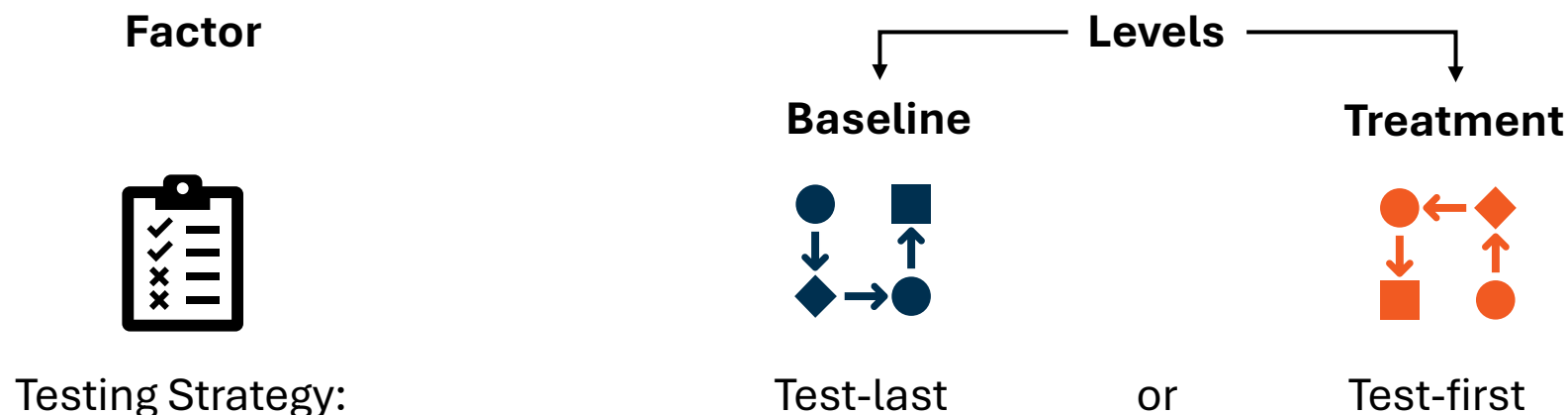


Oftentimes, one experiment involves multiple objects at once (e.g., source code, requirements specification, etc.)

Definition: Treatment

A controlled experiment is an empirical research method where we randomly assign a **treatment** to combinations of subjects and objects and observe its effect on one or more dependent variables.

Treatment: one particular value of a factor (an independent variable), opposed to the *baseline*



Definition: Dependent Variables

A controlled experiment is an empirical research method where we randomly assign a treatment to combinations of subjects and objects and observe its effect on one or more **dependent variables**.

Dependent variable: variable potentially affected by the independent variables (also called: response variable, outcome)



Time



Number of defects



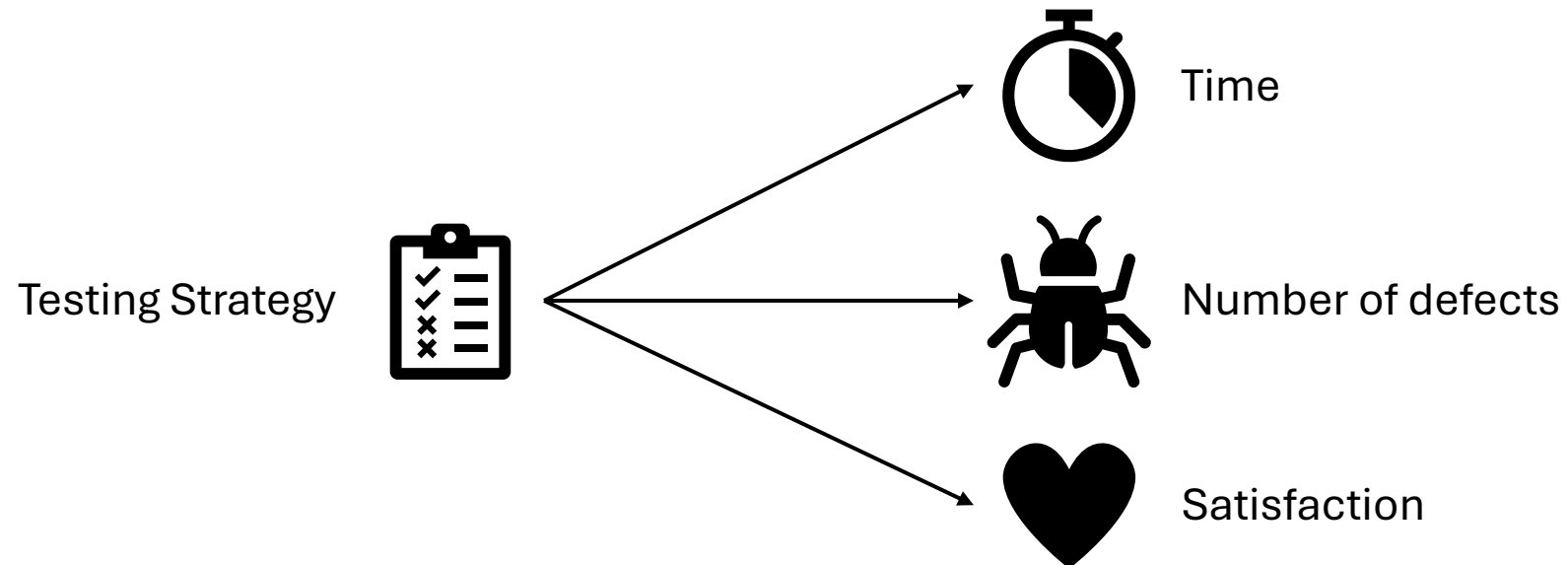
Satisfaction



One and the same treatment may have different effects on different outcome variables.

Definition: Dependent Variables

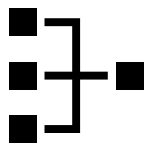
A controlled experiment is an empirical research method where we randomly assign a treatment to combinations of subjects and objects and **observe its effect** on one or more dependent variables.



Definition: Test

A controlled experiment is an empirical research method where we randomly **assign a treatment** to combinations of subjects and objects and observe its effect on one or more dependent variables.

Test: combination of a treatment, subject, and object

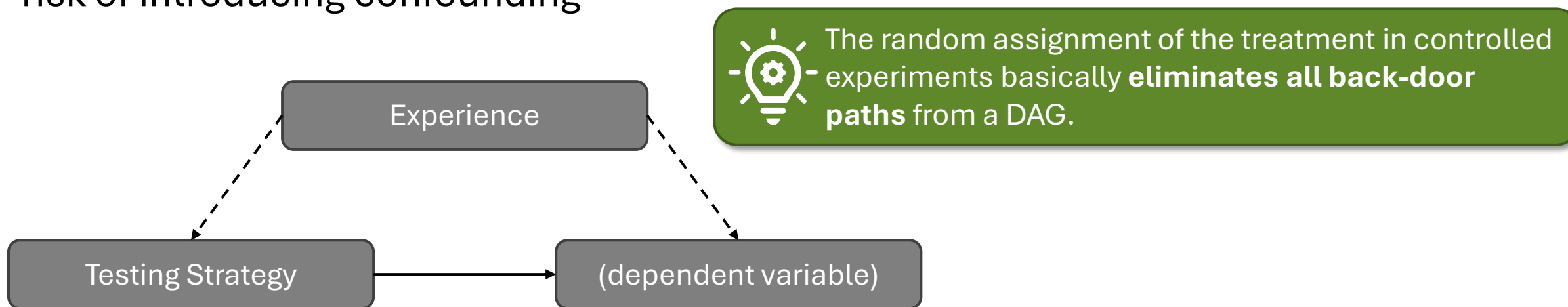


Person	System	Testing Strategy	Time	Number of defects	Satisfaction
P1	S1	 Test-last	1:45h	4	very high
P2	S1	 Test-first	1:12h	7	medium

Definition: Random

A controlled experiment is an empirical research method where we **randomly** assign a treatment to combinations of subjects and objects and observe its effect on one or more dependent variables.

Random assignment: without random assignment of treatments, we run the risk of introducing confounding





Experimentation Process

How to plan, design, prepare, execute, and analyze an experiment

Experimentation Process

Carrying out an experiment usually involves five steps:



Scoping



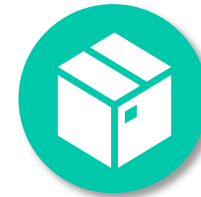
Planning



Operation



Analysis and
interpretation



Presentation
and package

Step 1: Scoping



During scoping, the researcher should **clearly state a hypothesis** and **define the objective and goals** of the experiment using the following framework:

- Objective of study (what is studied?),
- Purpose (what is the intention?),
- Quality focus (which effect is studied?),
- Perspective (whose view?), and
- Context (where is the study conducted?).

Step 2: Planning



Once the scope is defined, the researchers plan the experiment by preparing the environment, personnel, and material. This step usually involves

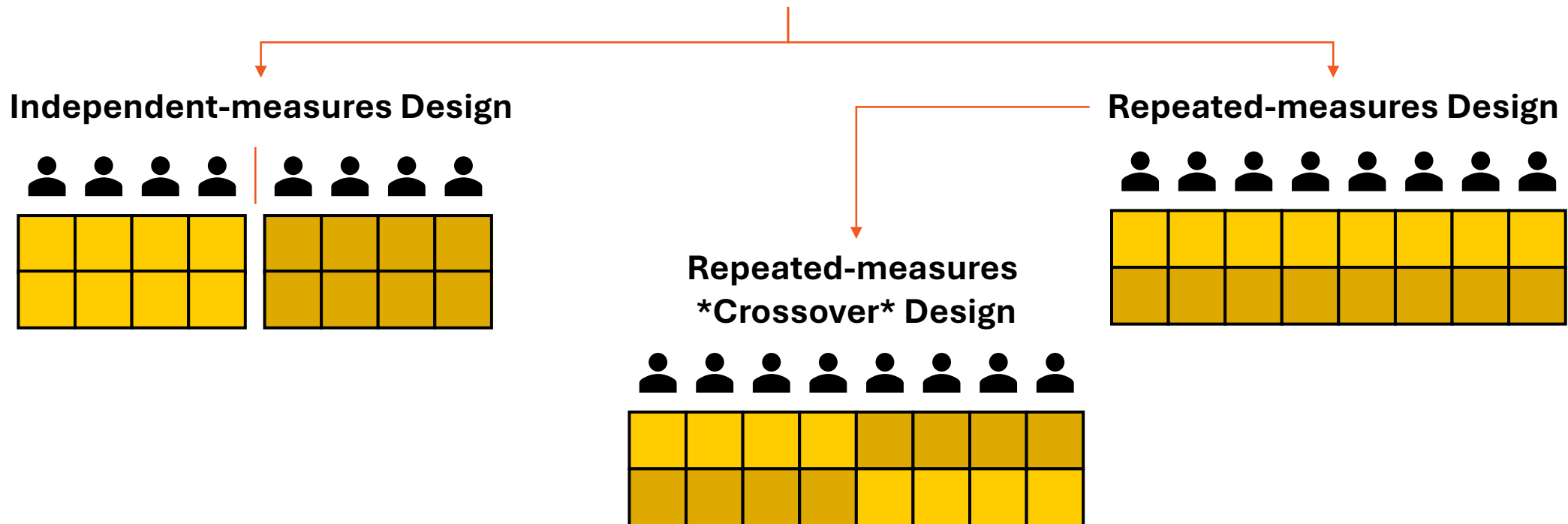
- Variable and context selection
- Selection of subjects
- Choice of design type
- Instrumentation
- Validity evaluation

Step 2: Planning



The **design** of an experiment determines, how subjects receive treatments.

Types of experimental Design



Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.

Step 3: Operation



Once the experiment is planned and scoped, the researcher can implement it, which usually consists of:

- Preparation (including recruitment),
- Execution, and
- Data validation

Step 3: Operation



The recruitment of participants seems trivial but has a detrimental effect on the validity of the experiment.

There are several sampling strategies to choose from:

- **Non-probability sampling:** convenience, purposive, referral-chain, ...
- **Probability sampling:** while frame, simple random, ...
- **Multi-stage sampling:** stratified, quota, cluster



The sampling strategy still does not answer: where do you get this sample from?

Step 4: Analysis and Interpretation



Once the researcher collected data, they can analyze it to derive answers about the initially posed questions. There are two types of data analysis:



Descriptive statistics

(e.g., bar charts, box plots)

Describe only the collected data



Inferential statistics

(e.g., hypothesis-tests, regression)

Infer properties of the larger target population from the sample



Due to the elimination of backdoor-paths, inferential statistics do not run the risk of confounding.

Step 4: Analysis of Crossover-Designs



Period 1	A	B
Period 2	B	A
	Seq. AB	Seq. BA

Step 4: Analysis of Crossover-Designs



Threats:

$$y_i = \alpha +$$

$$\beta_t t_i$$

+

$$\beta_p p_i$$

+

$$\beta_s s_i$$

+

$$\alpha_{pi}$$

+

$$\beta_c t_i \times s_i$$

**Addressing
Factor:**

Treatment

Period

Sequence

**Subject-specific
Random Effect**

**Treat. x Seq.
Interaction**

Period 1

A	B
B	A

Period 2

A	B
B	A

A	B
B	A

p1 p2	p3 p4
p1 p2	p3 p4

A	B
B	A

Seq. AB Seq. BA

ID	p	s	t	y
1	1	AB	A	14.2
1	2	AB	B	11.7

Vegas, S., Apa, C., & Juristo, N. (2015). Crossover designs in software engineering experiments: Benefits and perils. *IEEE Transactions on Software Engineering*, 42(2), 120-135.

Step 5: Presentation and Package



Science is useless when it is not communicated. Once results are obtained and interpreted, the researcher should produce at least two artifacts:

1. **Experiment report** (e.g., a scientific paper or technical report)
2. **Replication package** (i.e., a collection of all the material that allows to check and repeat the experiment)

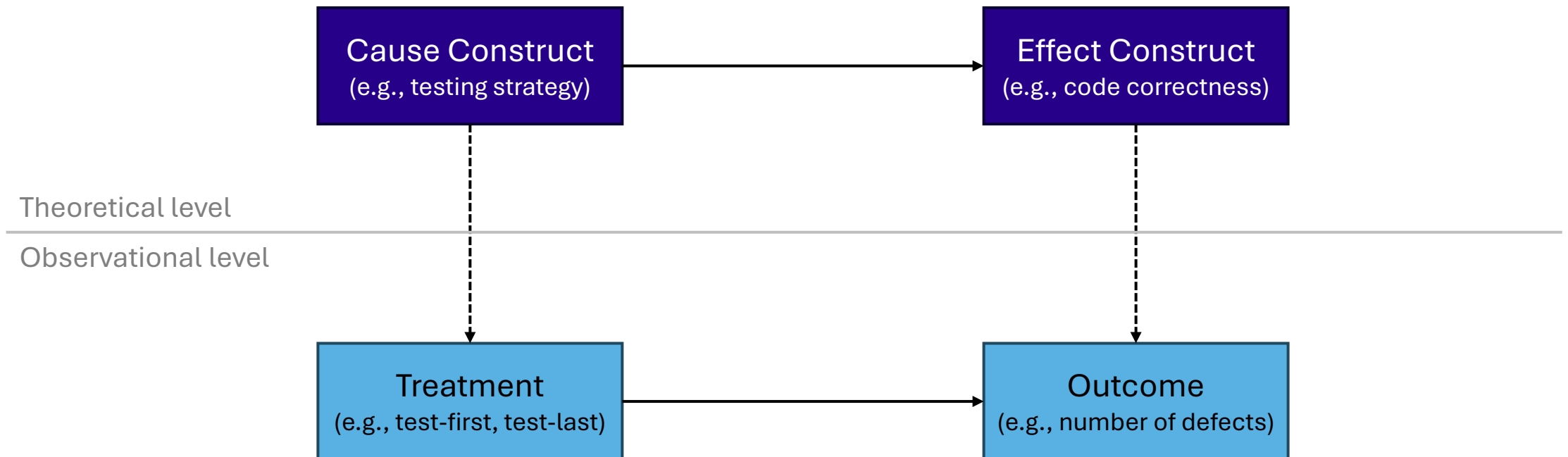
Optionally, disseminating the results via a poster, social media post, or infographics will increase its impact.



Validity

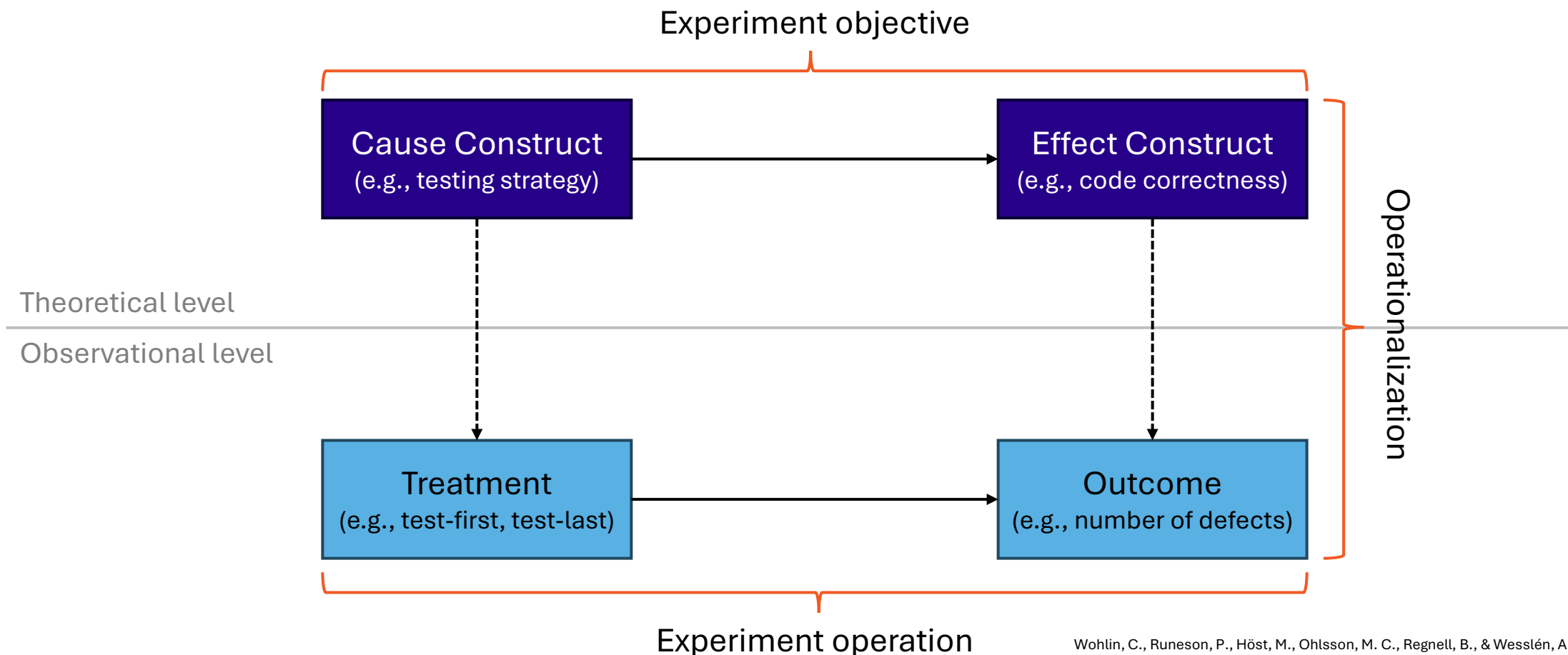
What makes a *good* experiment?

Validity of Experimental Results



Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.

Validity of Experimental Results

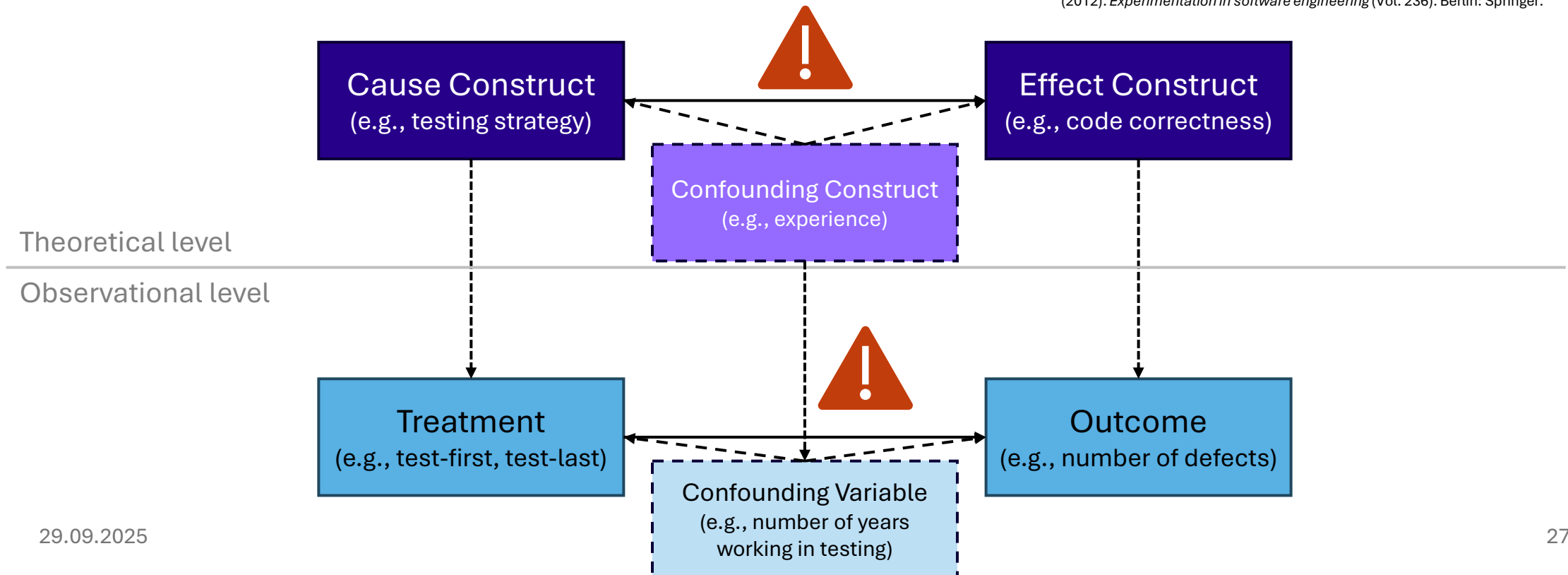


Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.

Internal validity

Threats to internal validity are influences that can affect the independent variable with respect to causality, without the researcher's knowledge.

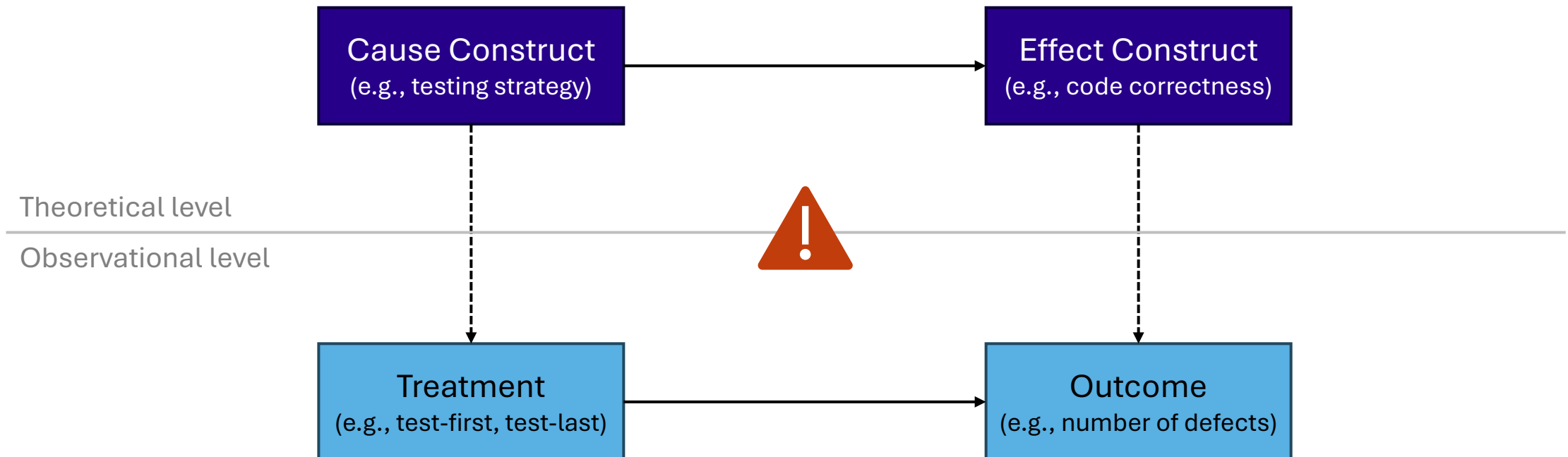
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.



External validity

Threats to external validity are conditions that limit our ability to generalize the results of our experiment to industrial practice.

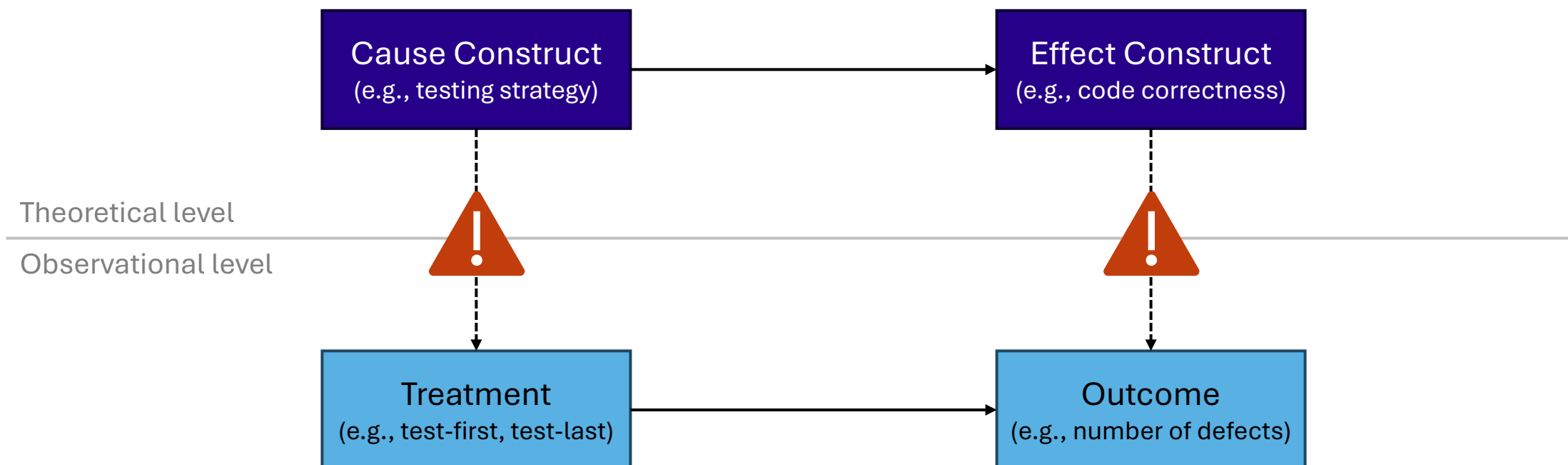
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.



Construct validity

Construct validity concerns generalizing the result of the experiment to the concept or theory behind the experiment.

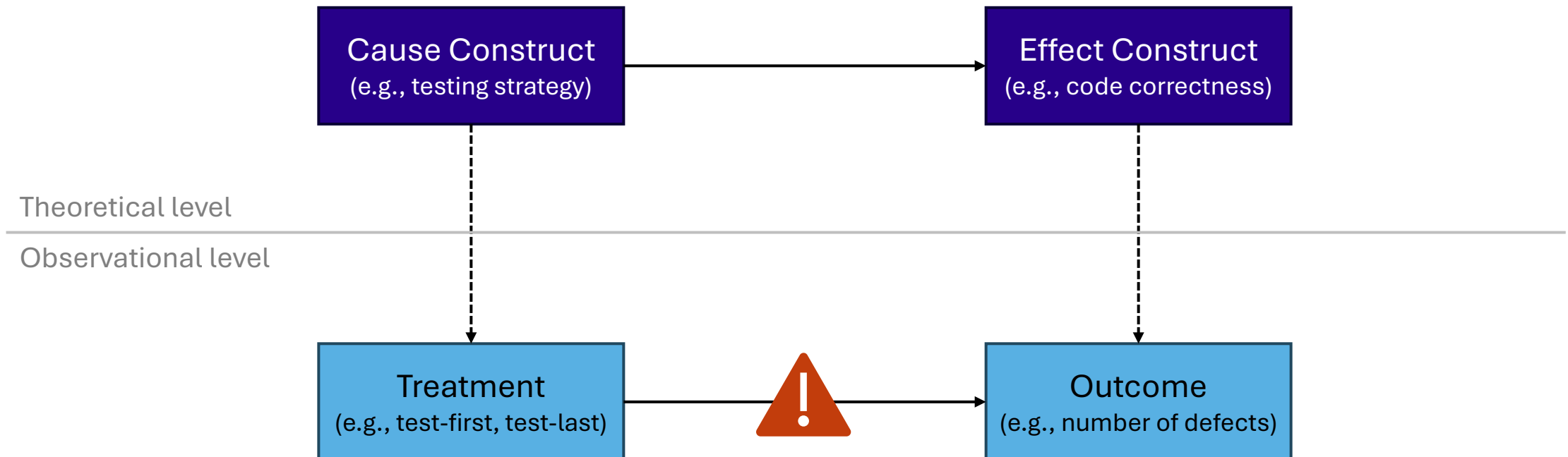
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.



Conclusion validity

Threats to the conclusion validity are concerned with issues that affect the ability to draw the correct conclusion about relations between the treatment and the outcome of an experiment.

Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.





Conclusion

Limitations,

Limitations

Experimentation is an inapplicable research method when randomly assigning treatments to subjects is



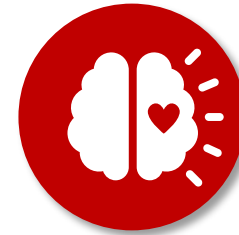
Expensive

Recruiting eligible
participants is difficult



Unrealistic

Experiments perturb the
natural context



Impossible

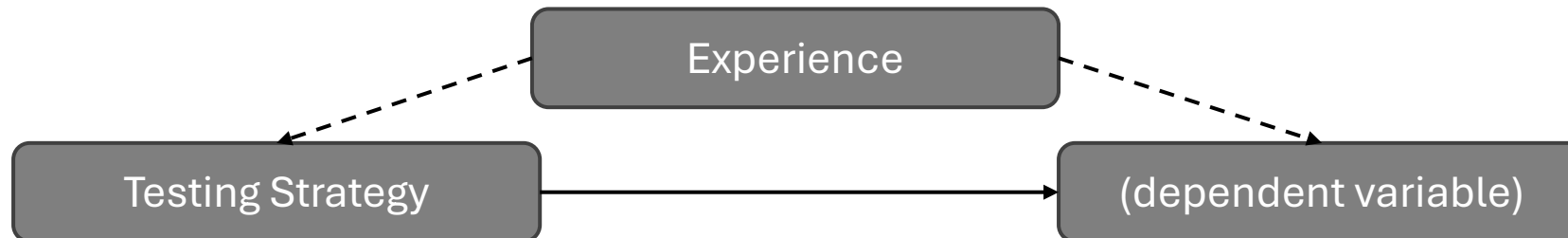
Some factors are
impossible to control

In these cases, we need to use alternative research methods.

Alternative: Observational Methods

Research methods studying the effect of an independent variable where the treatments are not randomly assigned are **observational methods** (e.g., case studies).

However, observational methods are prone to collect data where the effect of interest is confounded.



Because this is not uncommon, we require the methods for statistical causal inference.

Replication

Because a single experiment never achieves perfect validity, the scientific community needs to **replicate** it.



Schmidt S. Shall we really do it again? The powerful concept of replication is neglected in the social sciences. *Rev Gen Psychol.* 2009; 13(2): 90–100. <https://doi.org/10.1037/a0015108>

Credibility of scientific claims is established with evidence for their replicability using new data.

Exact replications are difficult to achieve and sometimes not even intended. Replications can vary across four dimensions:

1. **Operationalization:** treatment application and transmission, metrics, measurements, ...
2. **Population:** subject and object population
3. **Protocol:** experimental design, guides, data analysis techniques, ...
4. **Experimenter:** designer, trainer, monitor, measurer, analyst

Gómez, O. S., Juristo, N., & Vegas, S. (2014). Understanding replication of experiments in software engineering: A classification. *Information and Software Technology*, 56(8), 1033-1048.

The Big Picture

Empirical evidence gathered through experiments contributes to advancing reliable (i.e., valid) knowledge about causes and effects in a scientific domain (e.g., software engineering).



One single experiment will not change the world, but the accumulation and synthesis of many well-done experiments produces the knowledge about how to engineer software well.

References

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